5.11 Classwork: Trigonometric Functions, Identities, and Integration — Markscheme

1. $f(x) = \sin x, \quad 0 \le x \le 2\pi$.

(a) $\displaystyle\int_0^{\pi} \sin x \, \mathrm{d}x = \big[-\cos x\big]_0^{\pi}$ **(M1)(A1)**

$= -\cos\pi - (-\cos 0) = 1 + 1 = 2$ **(A1)**

(b) $\displaystyle\int_{\pi}^{2\pi} \sin x \, \mathrm{d}x = -2$ **(A1)**

(by symmetry, or $\big[-\cos x\big]_{\pi}^{2\pi} = -1 - 1 = -2$*)*

(c) $2 + (-2) = 0$. The definite integral measures *signed* area: R_1 contributes $+2$ and R_2 contributes -2; they cancel. **(R1)(R1)**

(d) Total area $= |2| + |-2| = 4$ **(A1)**

2. $f(x) = \cos^2 x$.

(a) $\cos 2x = 2\cos^2 x - 1$ **(M1)**

$\cos^2 x = \dfrac{\cos 2x + 1}{2} = \dfrac{1}{2}(1 + \cos 2x)$ **(AG)**

(b) $\displaystyle\int \cos^2 x \, \mathrm{d}x = \int \tfrac{1}{2}(1 + \cos 2x) \, \mathrm{d}x$ **(M1)**

$= \dfrac{x}{2} + \dfrac{\sin 2x}{4} + c$ **(A1)(A1)**

(c) $\left[\dfrac{x}{2} + \dfrac{\sin 2x}{4}\right]_0^{\pi/4}$ **(M1)**

$= \left(\dfrac{\pi}{8} + \dfrac{\sin(\pi/2)}{4}\right) - 0 = \dfrac{\pi}{8} + \dfrac{1}{4}$ **(A1)**

$= \dfrac{\pi + 2}{8}$ **(A1)**

3. $g(x) = \sin^2 x, \quad h(x) = \cos^2 x$.

(a) $\sin^2\dfrac{\pi}{4} = \left(\dfrac{\sqrt{2}}{2}\right)^2 = \dfrac{1}{2}$ **(A1)**

$\cos^2\dfrac{\pi}{4} = \left(\dfrac{\sqrt{2}}{2}\right)^2 = \dfrac{1}{2}$ **(A1)**

Both equal $\frac{1}{2}$, so the graphs intersect at $x = \frac{\pi}{4}$. ✓

(b) $\cos 2x = 1 - 2\sin^2 x$ **(M1)**

$$\sin^2 x = \frac{1 - \cos 2x}{2} = \frac{1}{2}(1 - \cos 2x)$$ **(AG)**

(c) $\displaystyle\int_0^\pi \sin^2 x \, \mathrm{d}x = \int_0^\pi \tfrac{1}{2}(1 - \cos 2x) \, \mathrm{d}x$ **(M1)**

$$= \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^\pi$$ **(A1)(A1)**

$$= \left(\frac{\pi}{2} - \frac{\sin 2\pi}{4}\right) - 0 = \frac{\pi}{2}$$ **(A1)**

(d) $\sin^2 x + \cos^2 x = 1$, so $\displaystyle\int_0^\pi \cos^2 x \, \mathrm{d}x = \int_0^\pi (1 - \sin^2 x) \, \mathrm{d}x$ **(M1)**

$$= \pi - \frac{\pi}{2} = \frac{\pi}{2}$$ **(A1)**

4. $f(x) = 4\sin x \cos x$.

(a) $f(x) = 4\sin x \cos x = 2(2\sin x \cos x) = 2\sin 2x$ **(M1)(AG)**

(b) Period $= \dfrac{2\pi}{2} = \pi$ **(A1)**

(c) Area $= \displaystyle\int_0^{\pi/2} 2\sin 2x \, \mathrm{d}x$ **(M1)**

$= \big[-\cos 2x\big]_0^{\pi/2}$ **(A1)(A1)**

$= -\cos \pi - (-\cos 0) = 1 + 1 = 2$ **(A1)**

(d) By symmetry the negative arch from $\frac{\pi}{2}$ to π has the same shape. **(R1)**

Total area $= 2 + 2 = 4$ **(A1)**

(e) $f'(x) = 4\cos 2x$ **(A1)**

$f'(x) = 0 \Rightarrow \cos 2x = 0 \Rightarrow 2x = \frac{\pi}{2} \Rightarrow x = \frac{\pi}{4}$ **(A1)**